Kalman Filter Theory and Applications Equation Drilldown

https://github.com/musicarroll/kalman_course

Michael L. Carroll

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Part I

The Five Basic Kalman Equations

Topics

- Understanding the Equations: Heuristic Introduction

- Understanding the Equations: Heuristic Introduction
- **Equation Drilldown: Taking the Equations Apart**

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- **Equation Drilldown: Taking the Equations Apart**
- State Space Concepts

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Equation Drilldown: Taking the Equations Apart

Topics

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- Mathematical Formulation of the Problem

Equation Drilldown: Taking the Equations Apart

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- Drilldown on State Dynamics and Covariance Extrapolation Equations

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- Exercises

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- **Answers** to Exercises

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- **Answers** to Exercises

Answers to Exercises

Resistor Revisited

Answers to Exercises

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1. Work through the resistor example by hand using $\hat{x}(0) = 100\Omega$, $\sigma = 0.1$, and $P(0) = 0$. Start the recursive loop with the gain equation first. Run through the loop three times.

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     and thus the new measurement model is:

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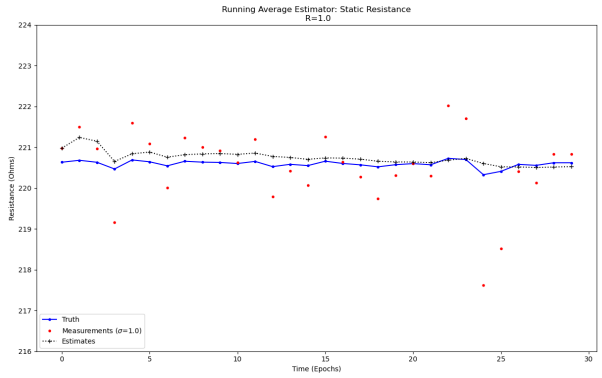
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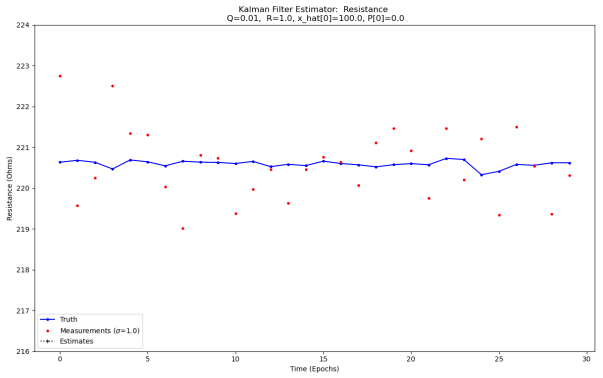
Resistor Revisited

Answers to Exercises

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(a) Simple Running Average Estimator

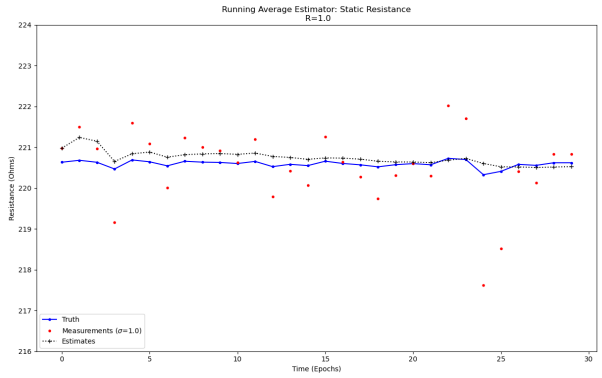


(b) Kalman Estimator

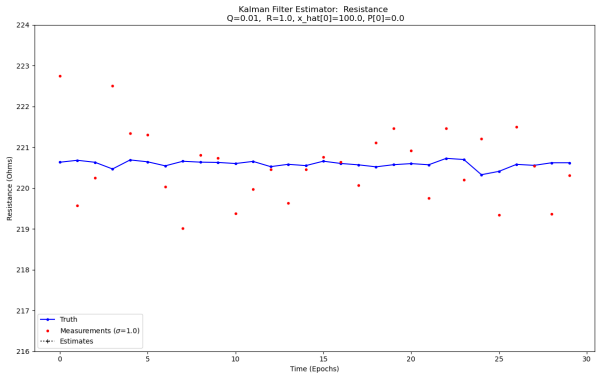
Figure: Comparing Running Average and Kalman Estimators **with process noise!!!**

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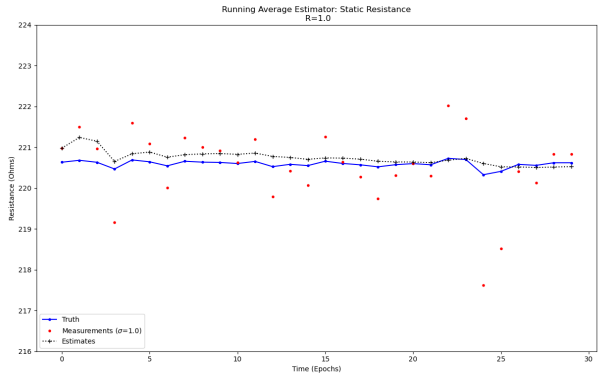
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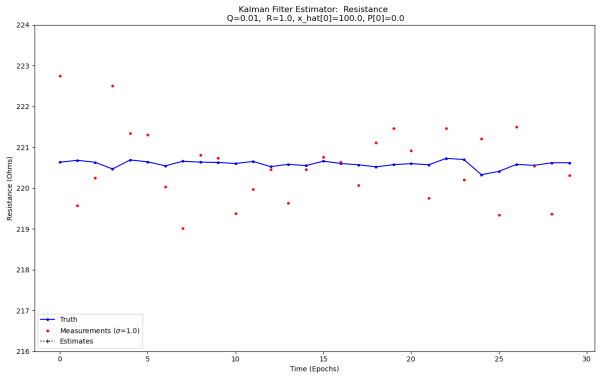
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$$K = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \left(\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \sigma^2 \right)^{-1}$$

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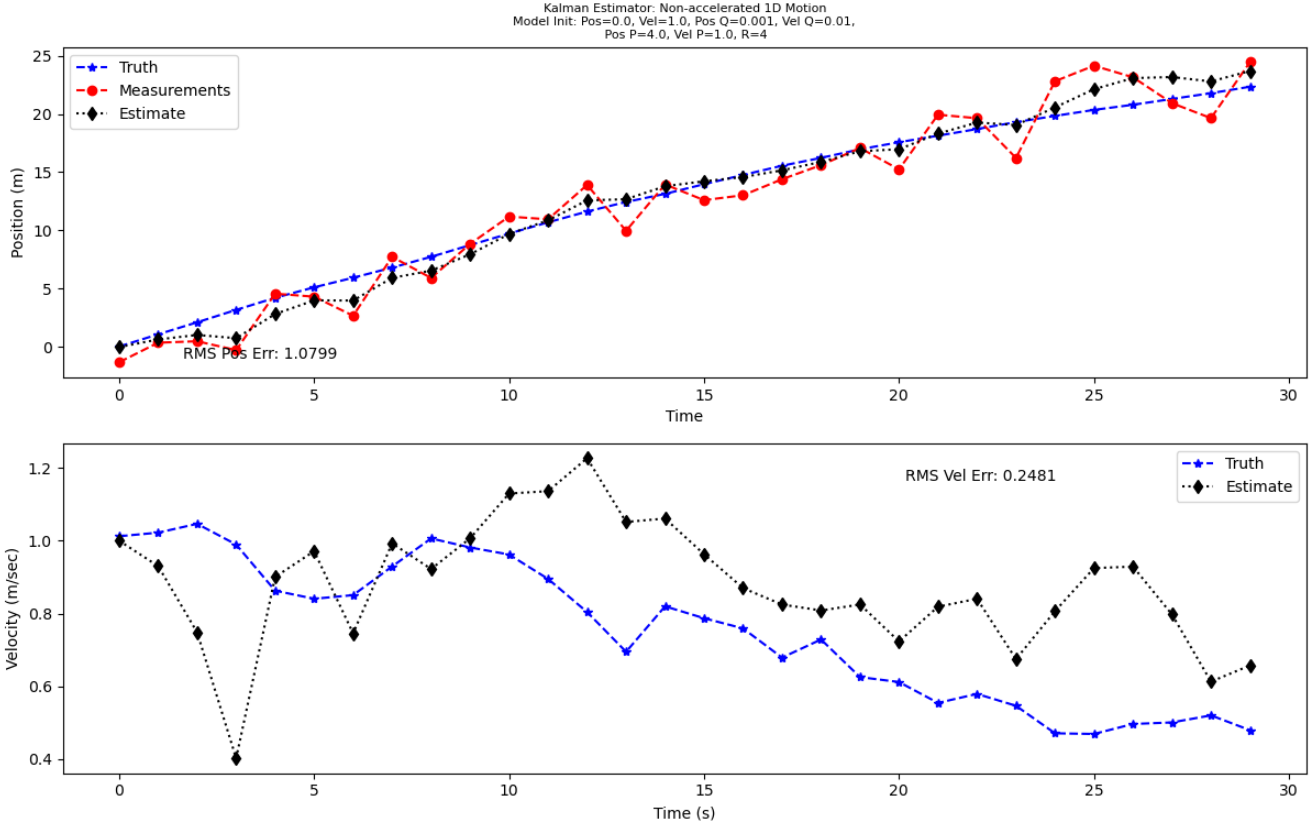


Figure: TME Plots Without Velocity Sensor

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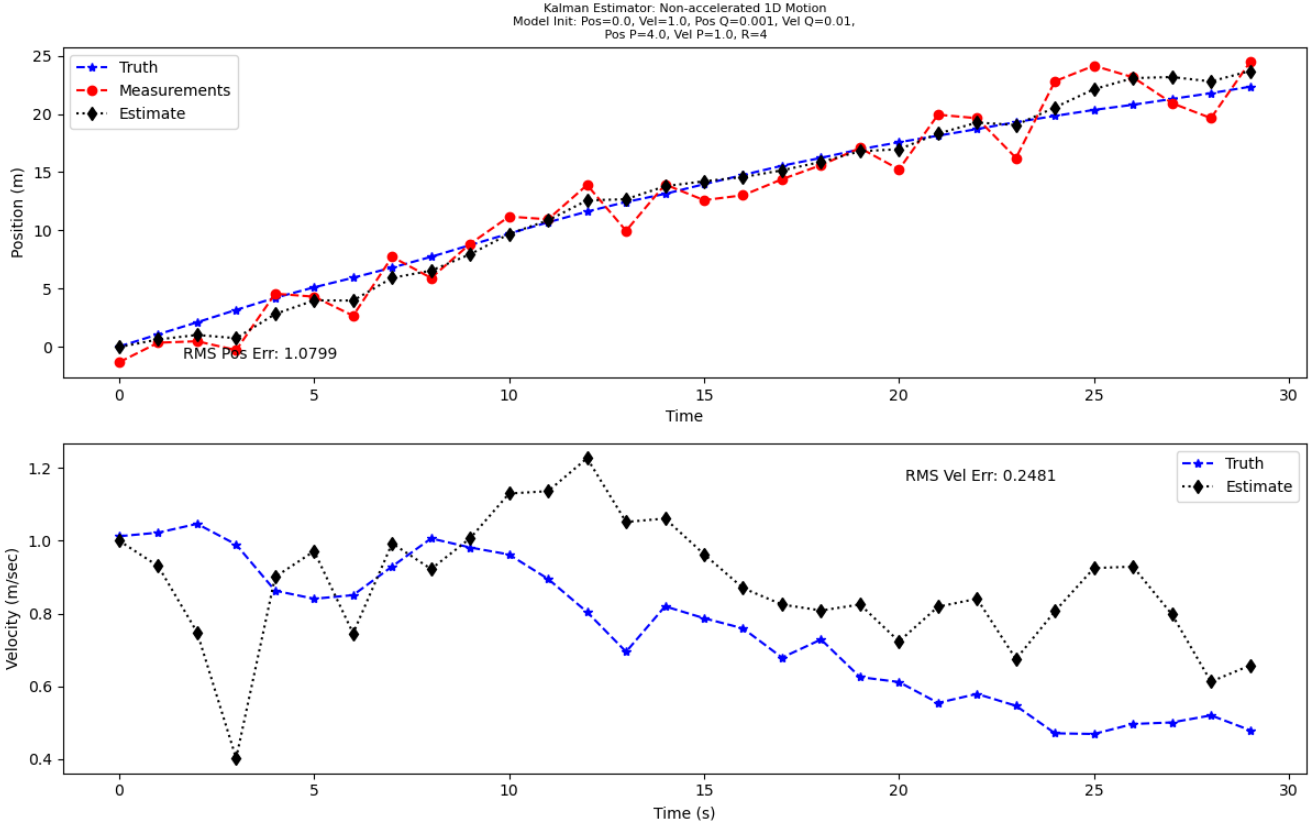


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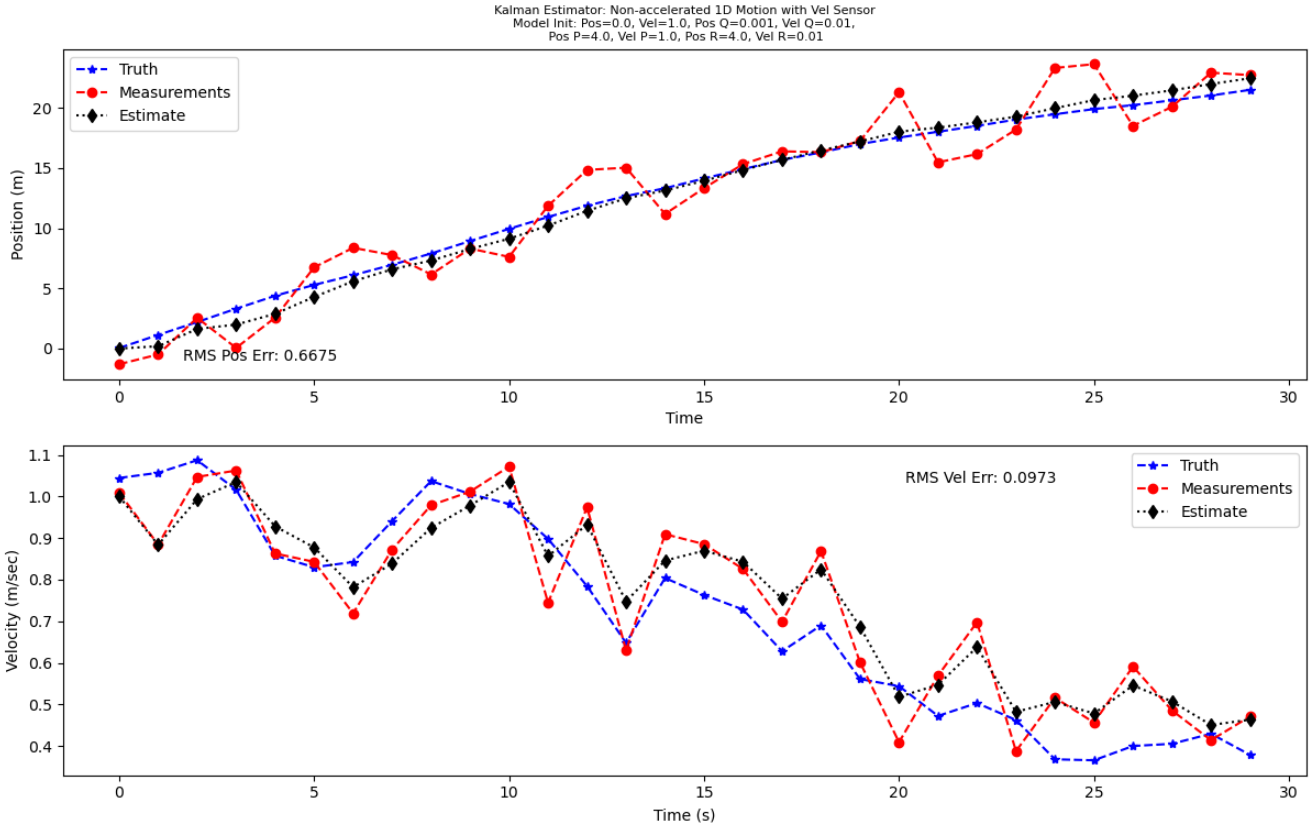


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Answers to Exercises

Constant Velocity Motion: TME Plots w/ Vel Sensor

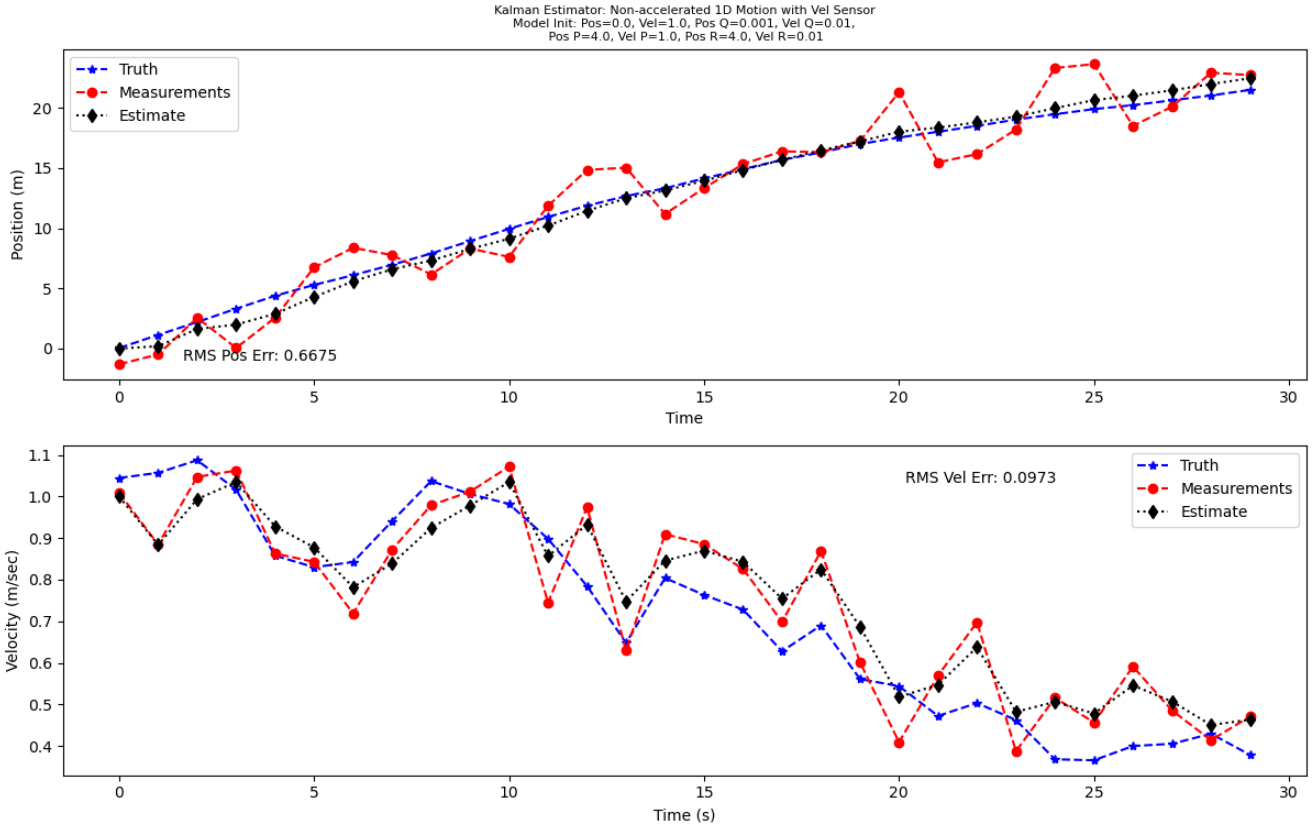


Figure: TME Plots With Velocity Sensor

Answers to Exercises

Constant Velocity Motion: Pos Sawtooth w/o Vel Sensor

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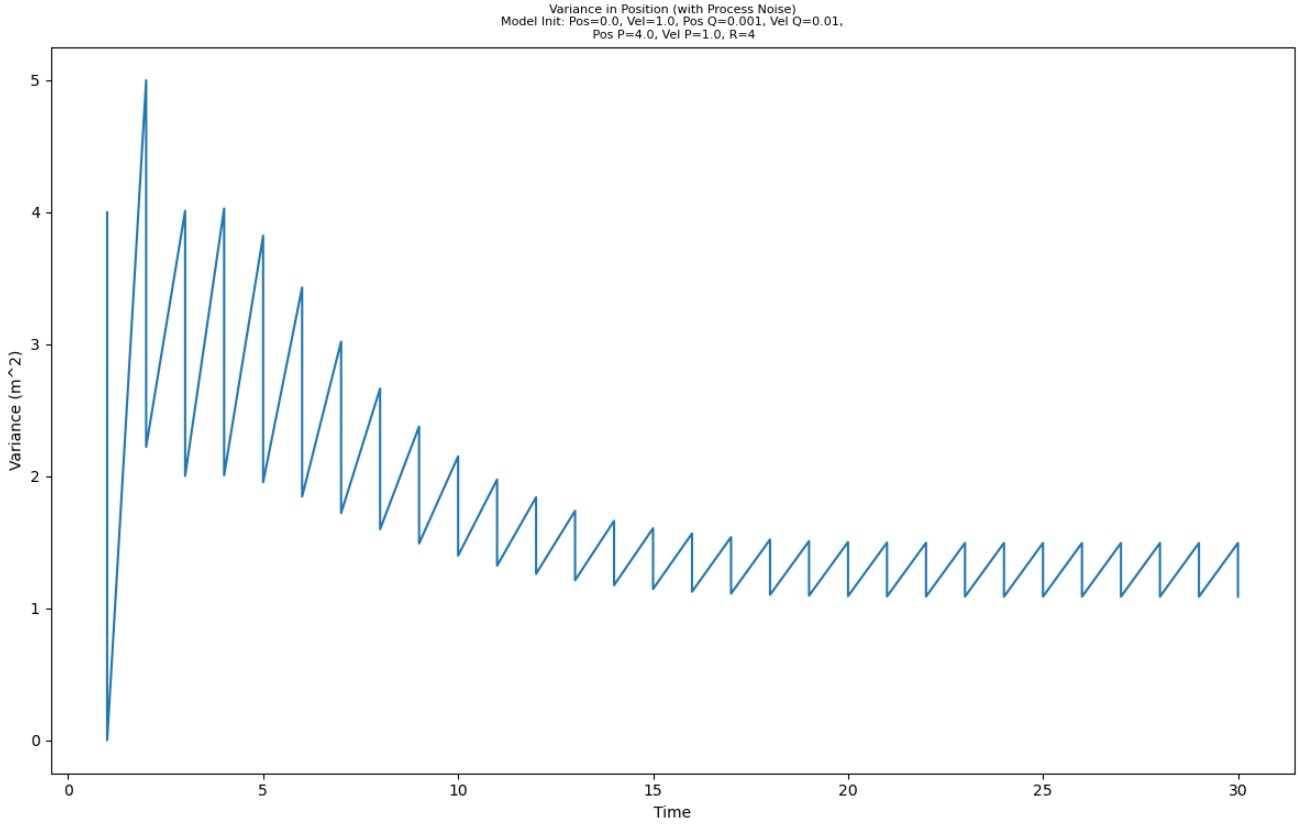


Figure: Position Sawtooth **Without** Velocity Sensor

Answers to Exercises

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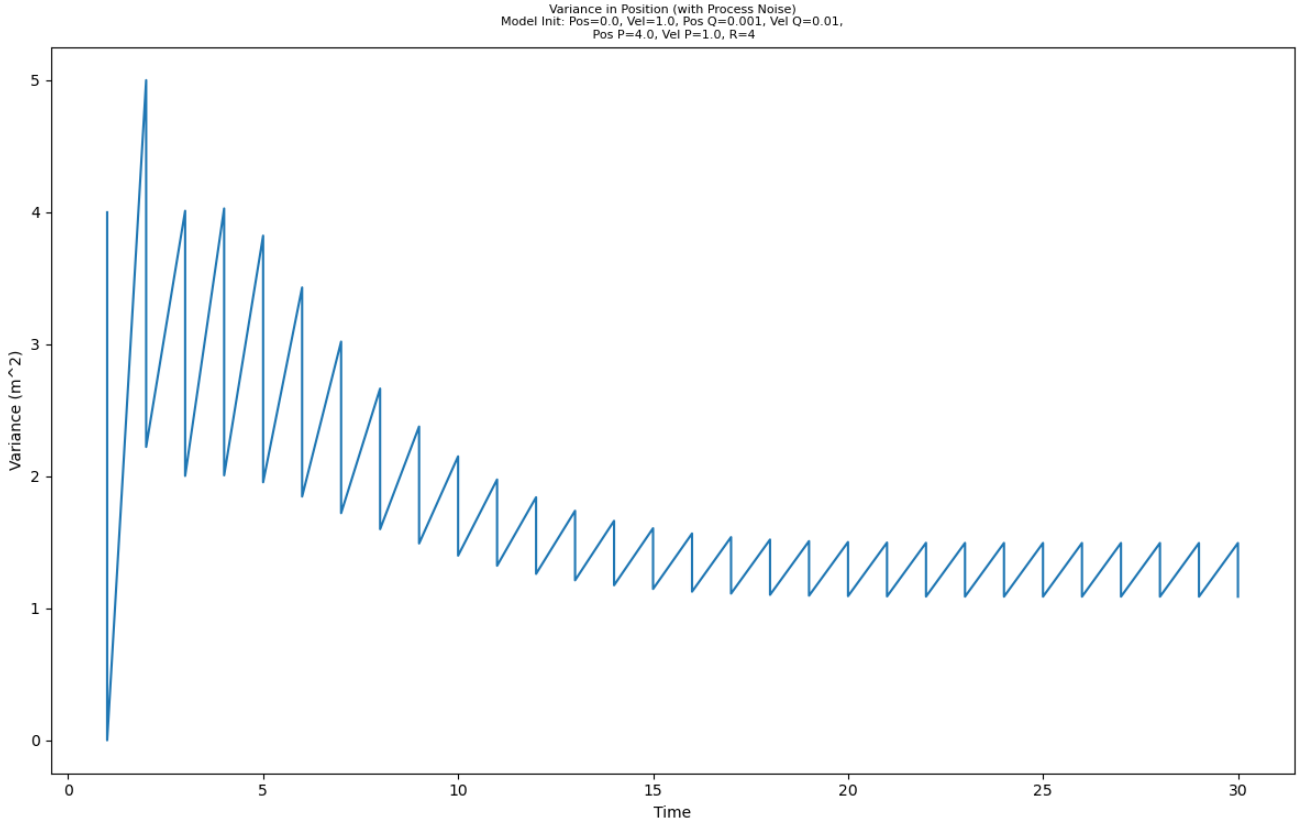


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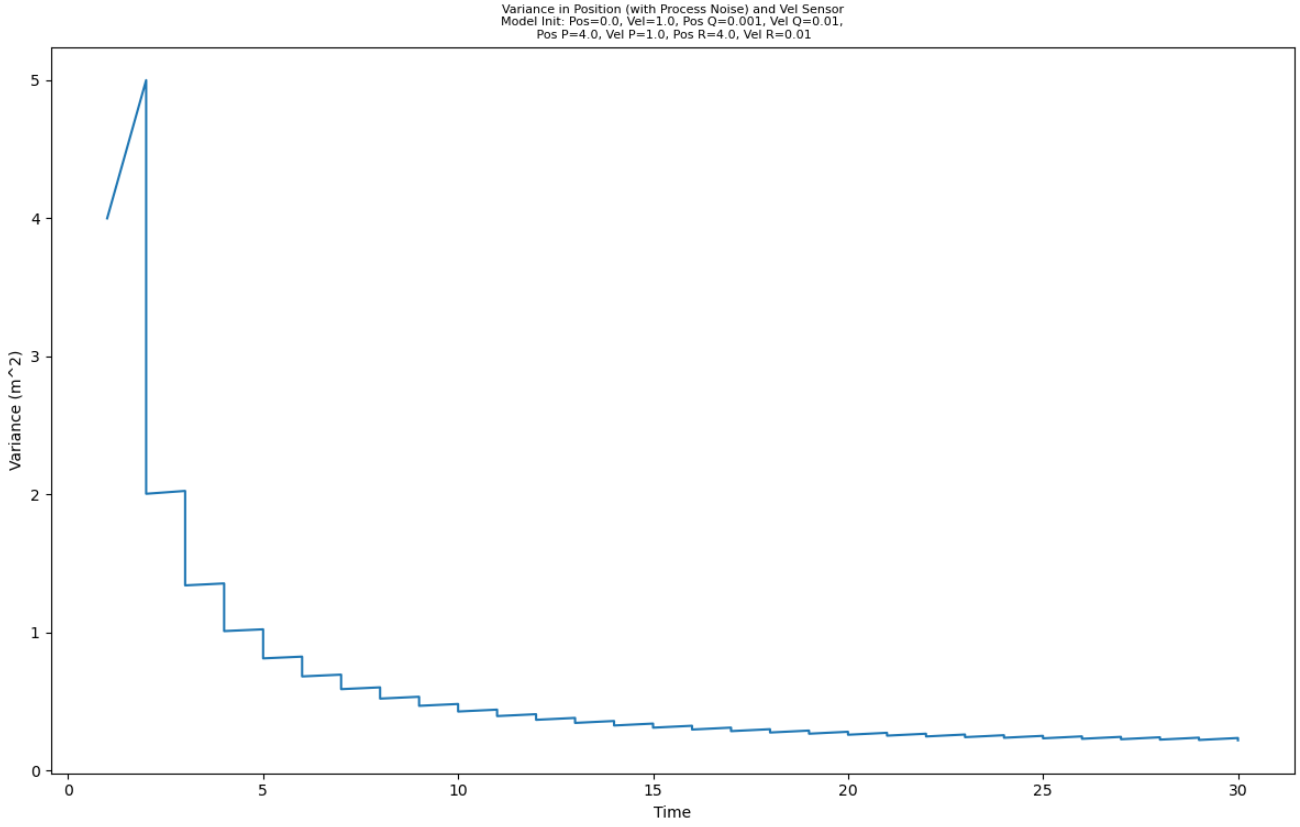


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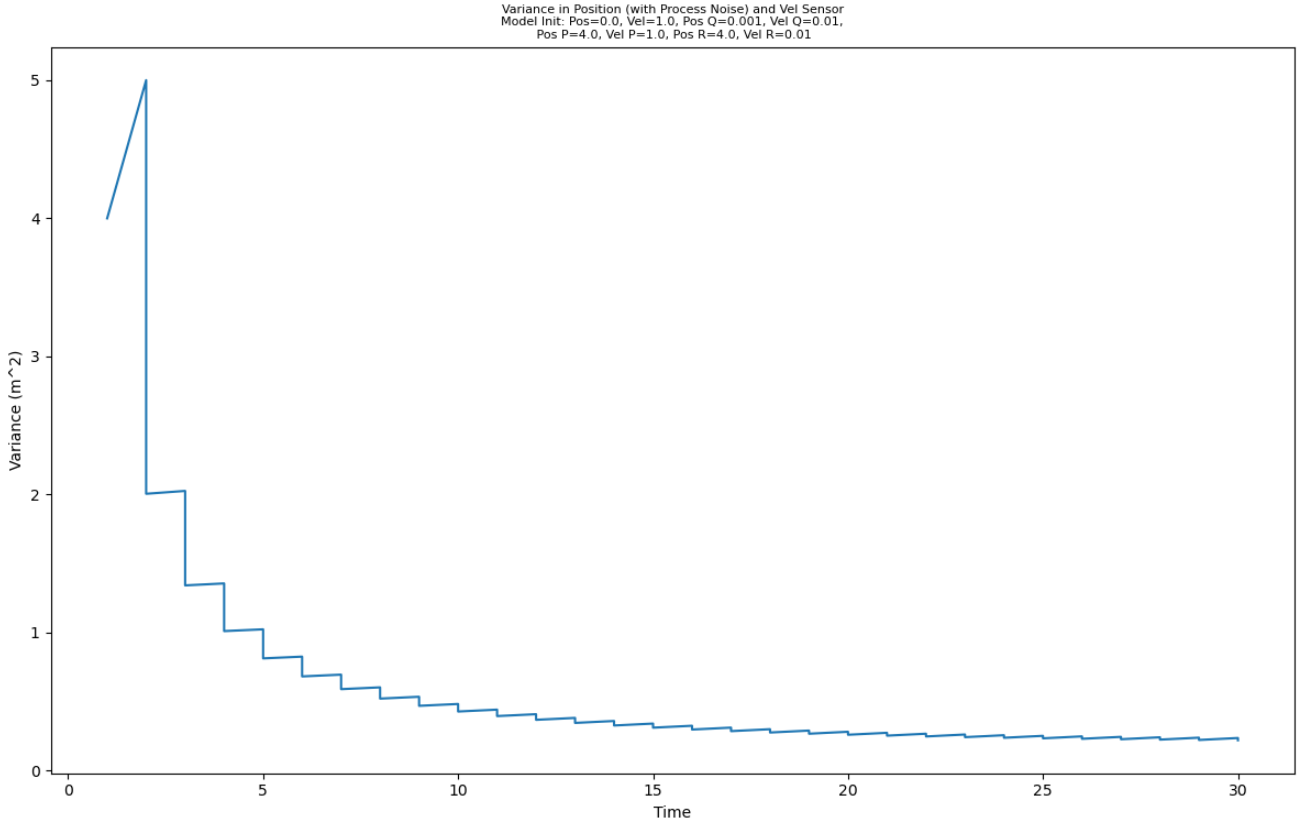


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Answers to Exercises

Constant Velocity Motion: Vel Sawtooth w/o Vel Sensor

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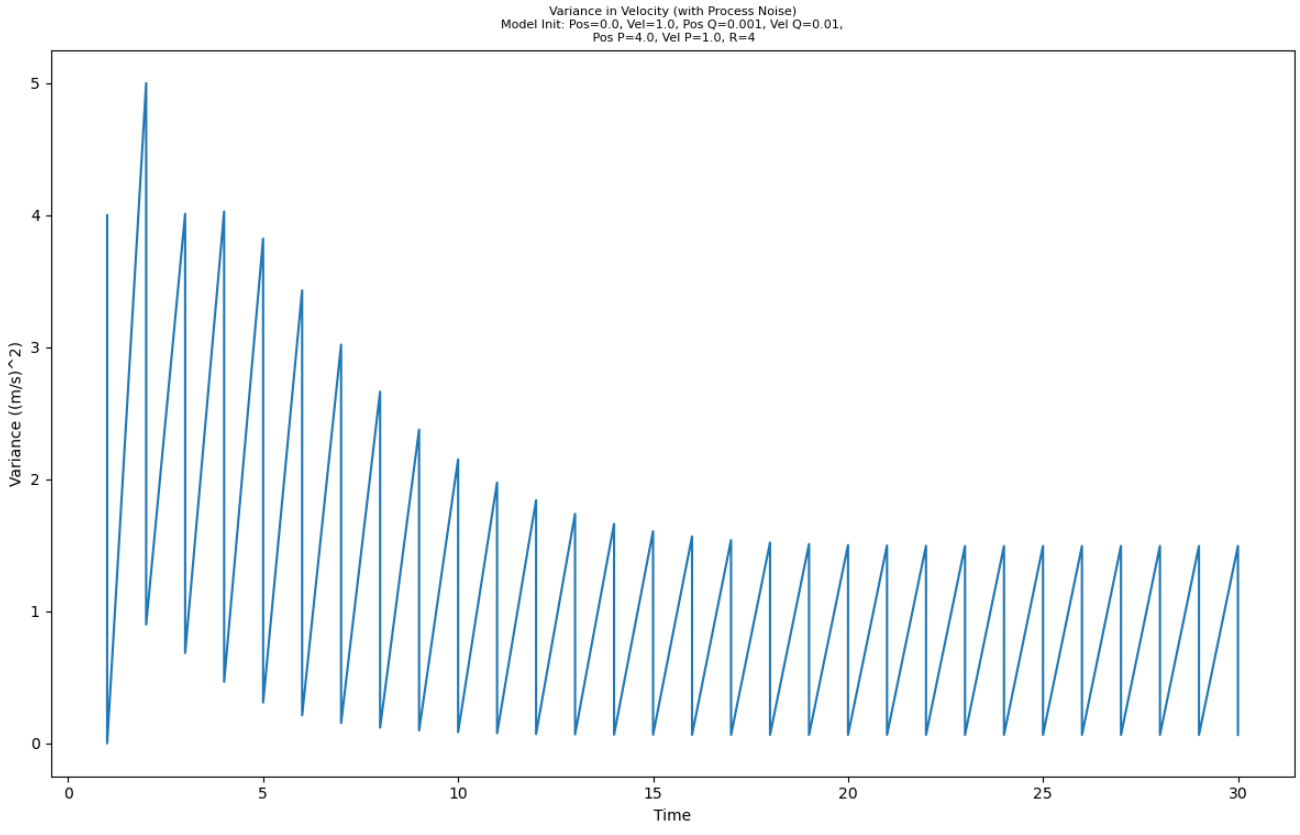


Figure: Velocity Sawtooth **Without** Velocity Sensor

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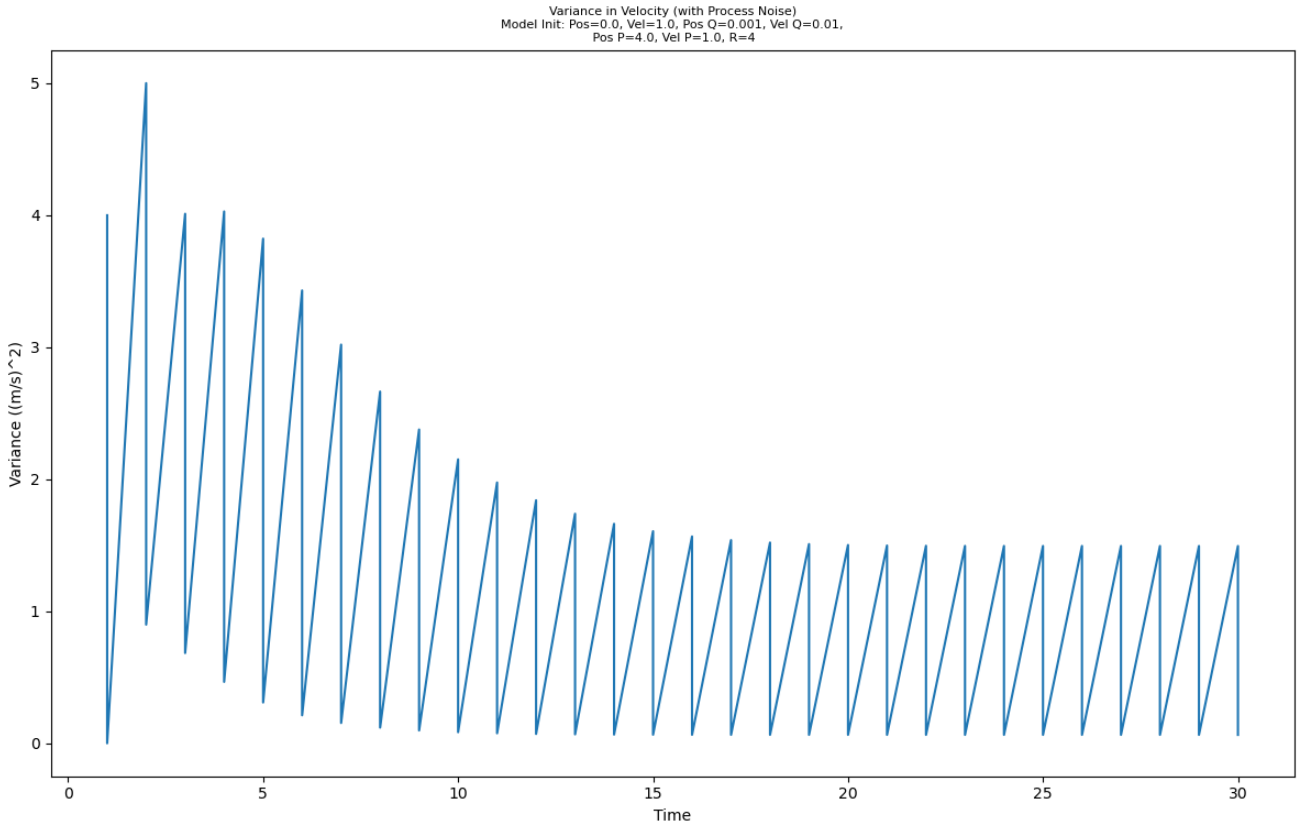


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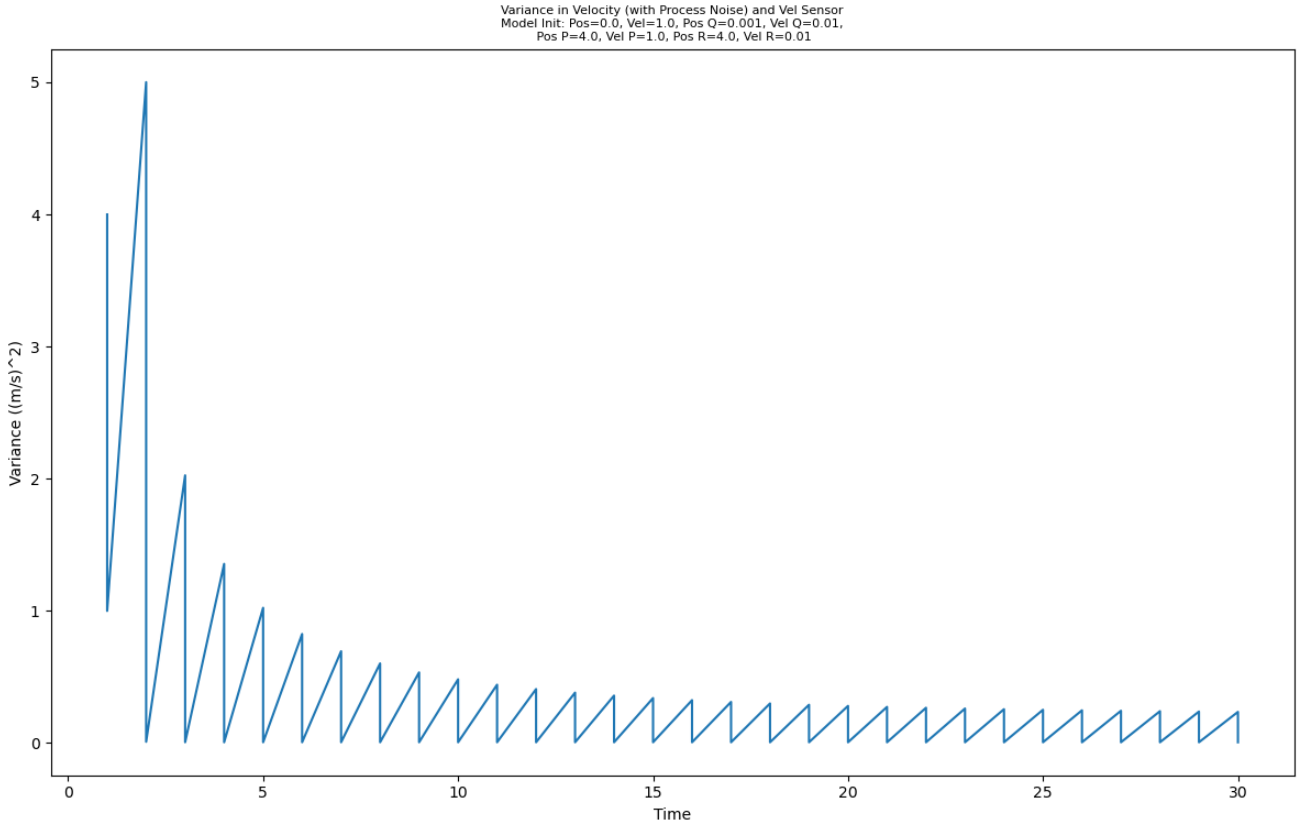


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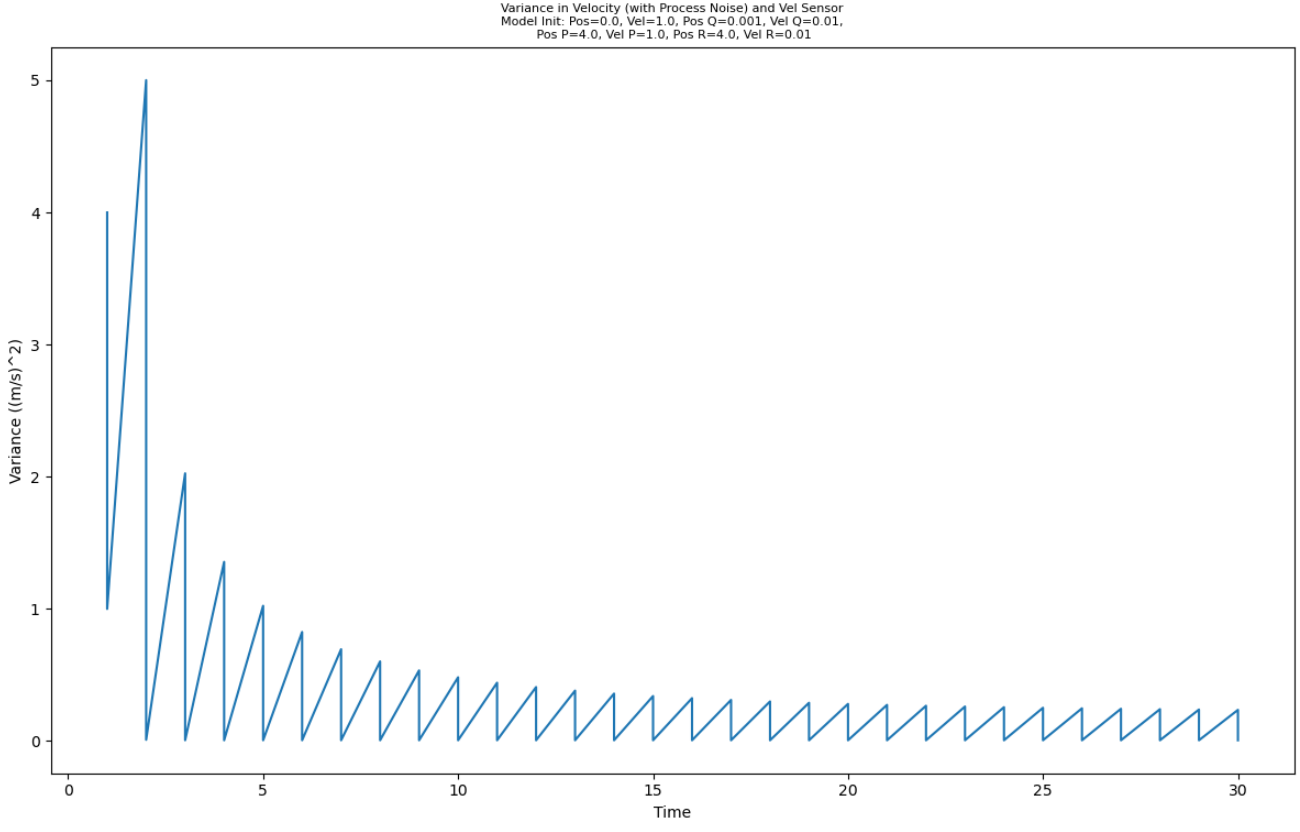


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Answers to Exercises

Damped Harmonic Oscillator

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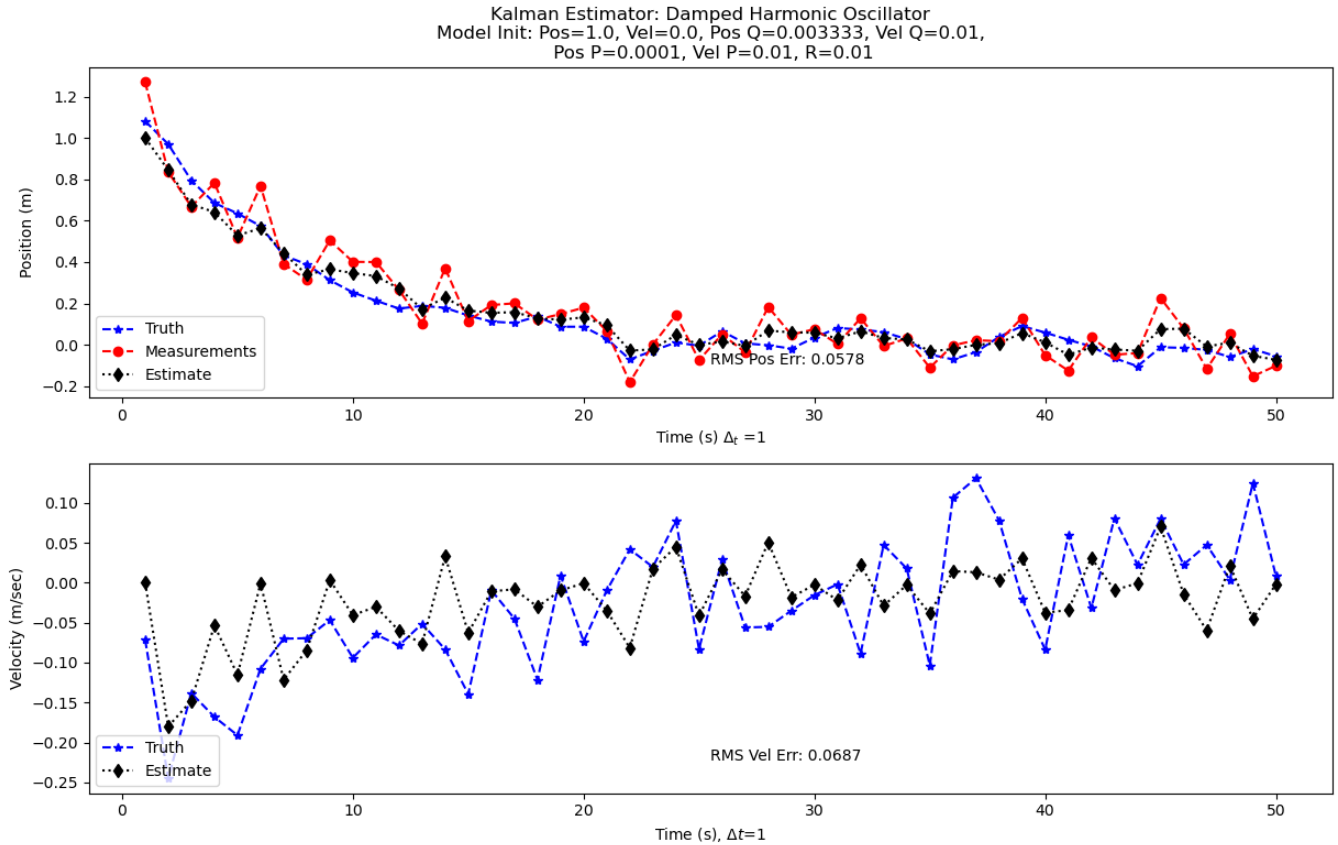


Figure: TME Plot w/o Vel Sensor

Answers to Exercises

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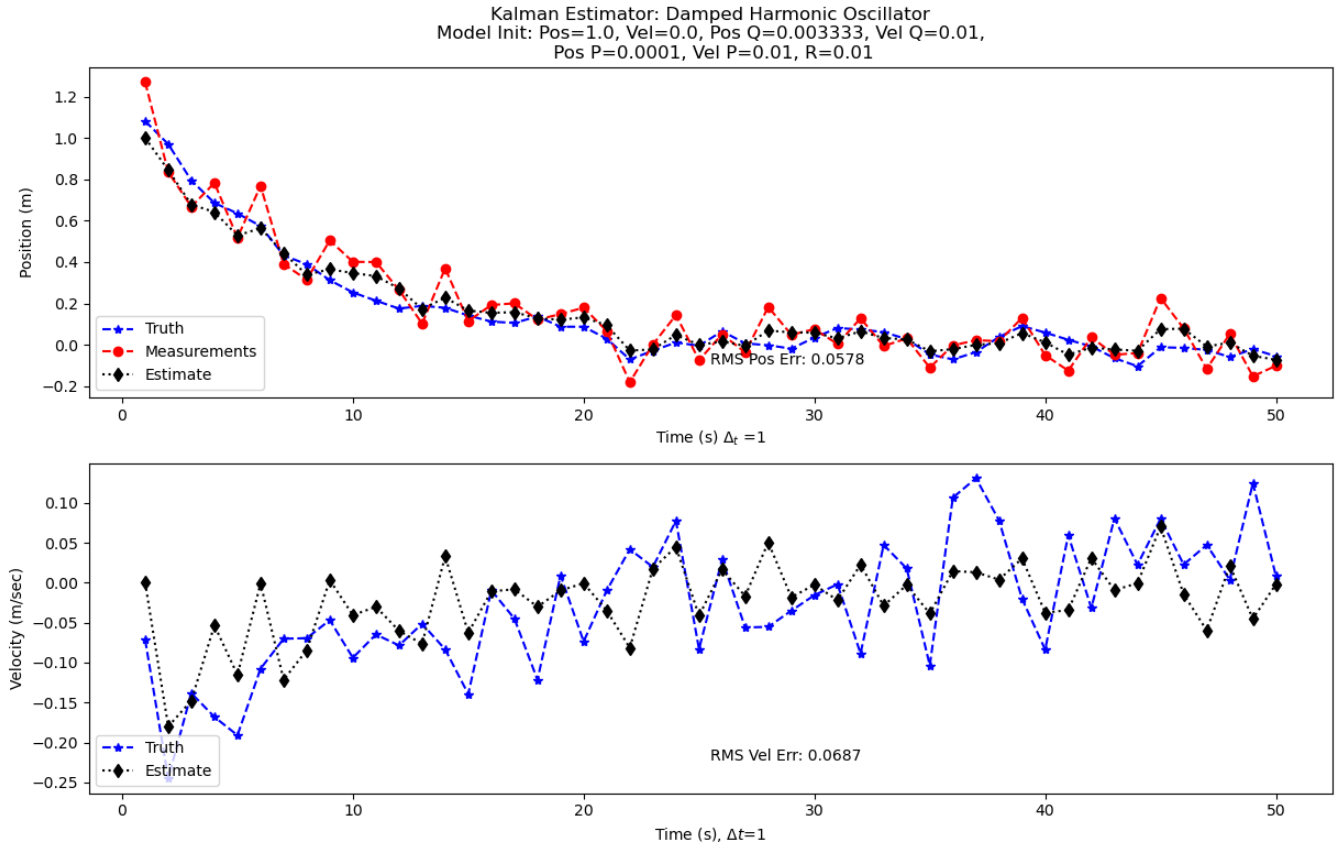


Figure: TME Plot w/o Vel Sensor

Answers to Exercises

Damped Harmonic Oscillator: TME Plot w/ Pos and Vel Sensors

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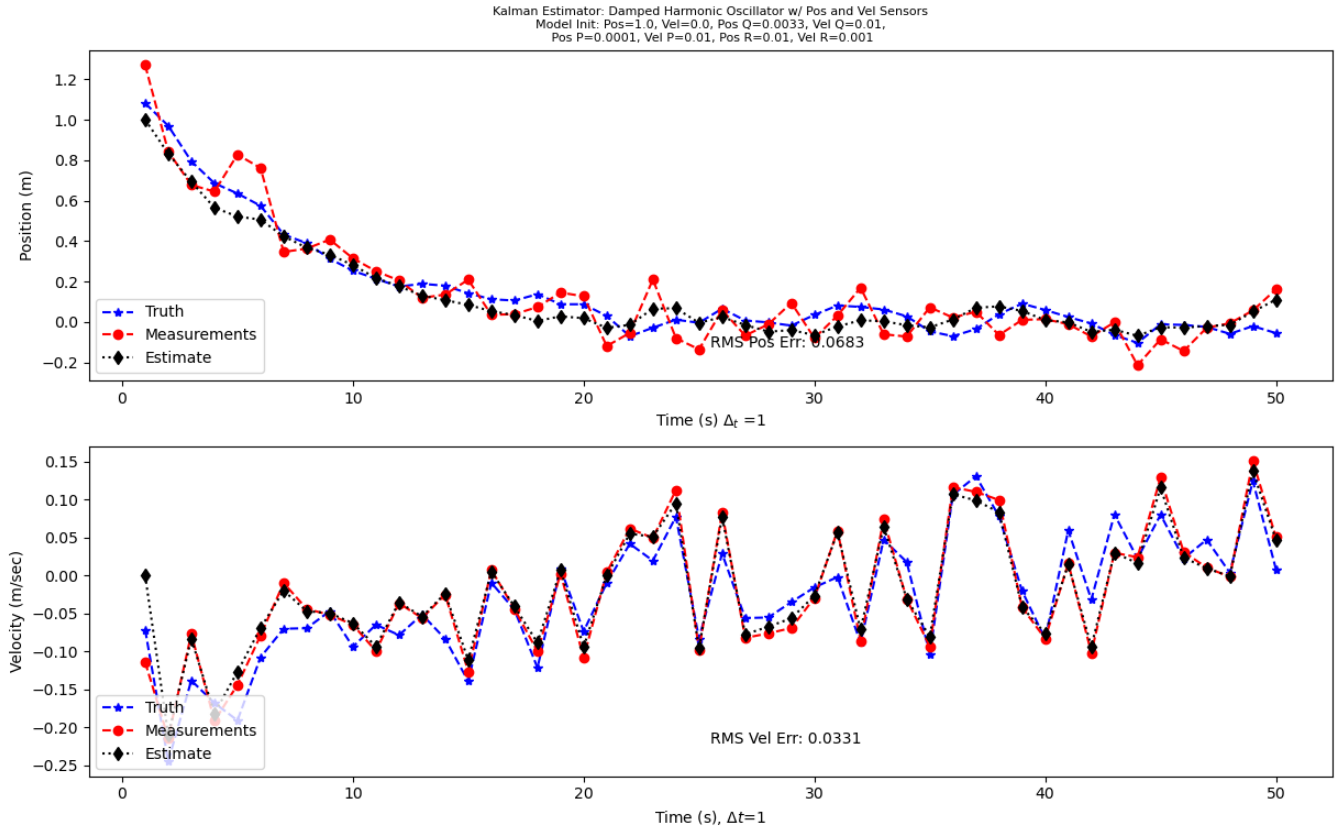


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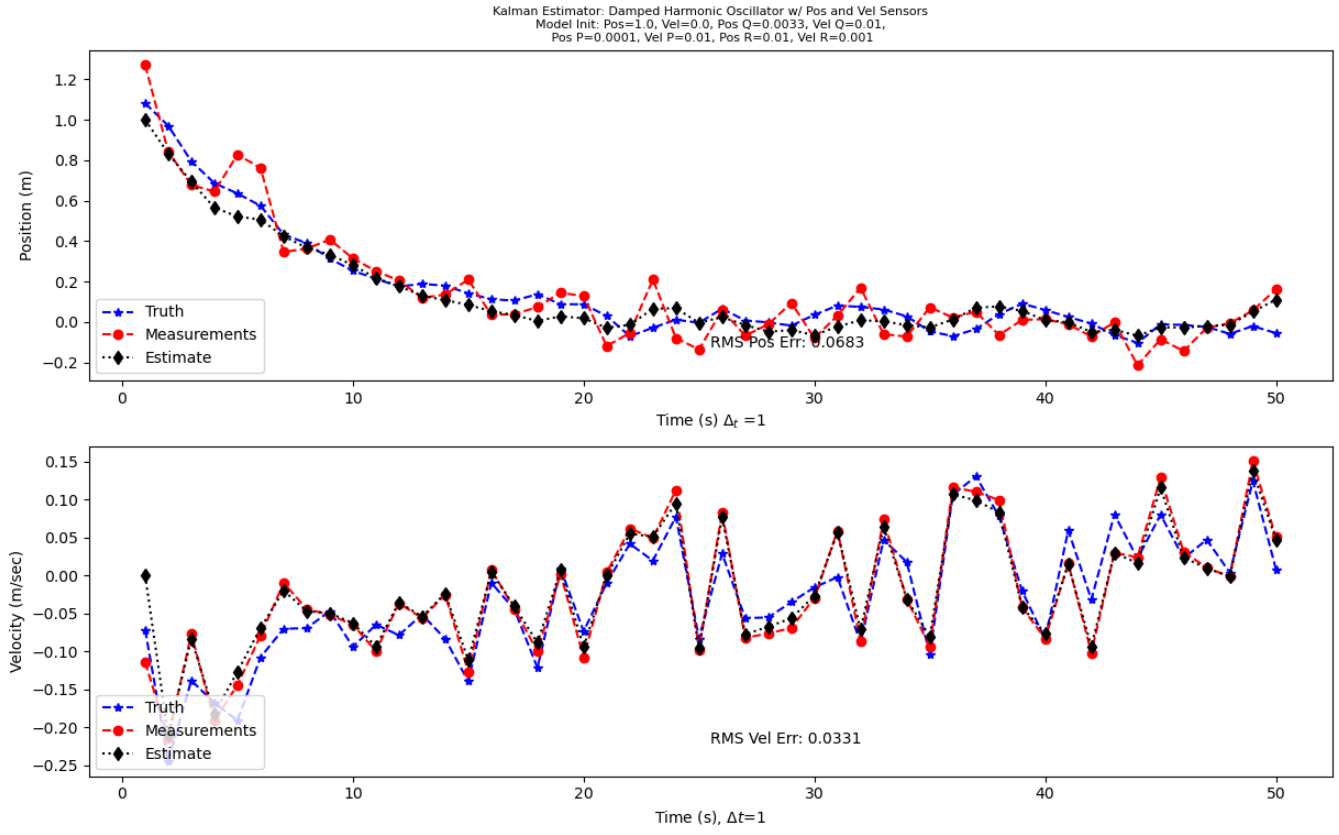


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Answers to Exercises

Damped Harmonic Oscillator: Position Variance Sawtooth w/o Vel Sensor

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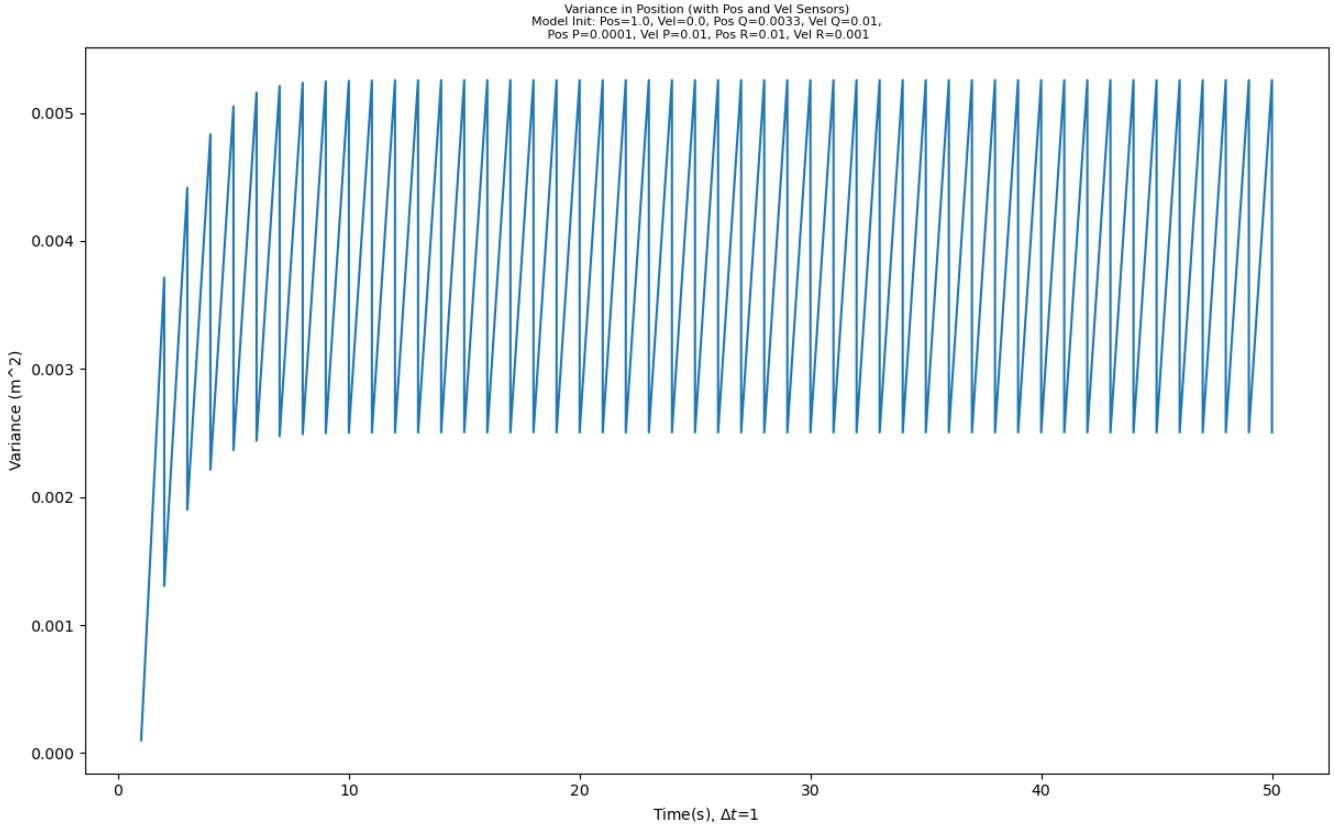


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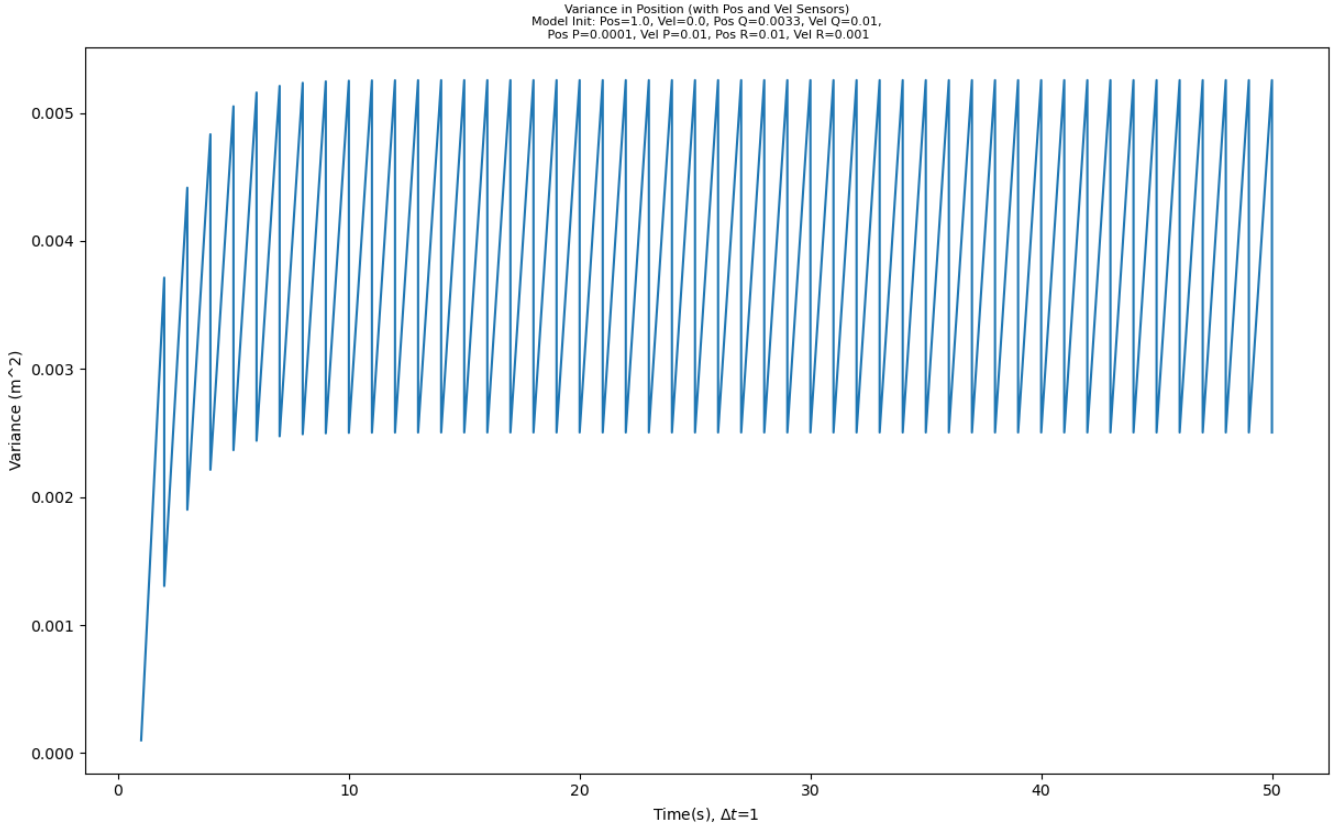


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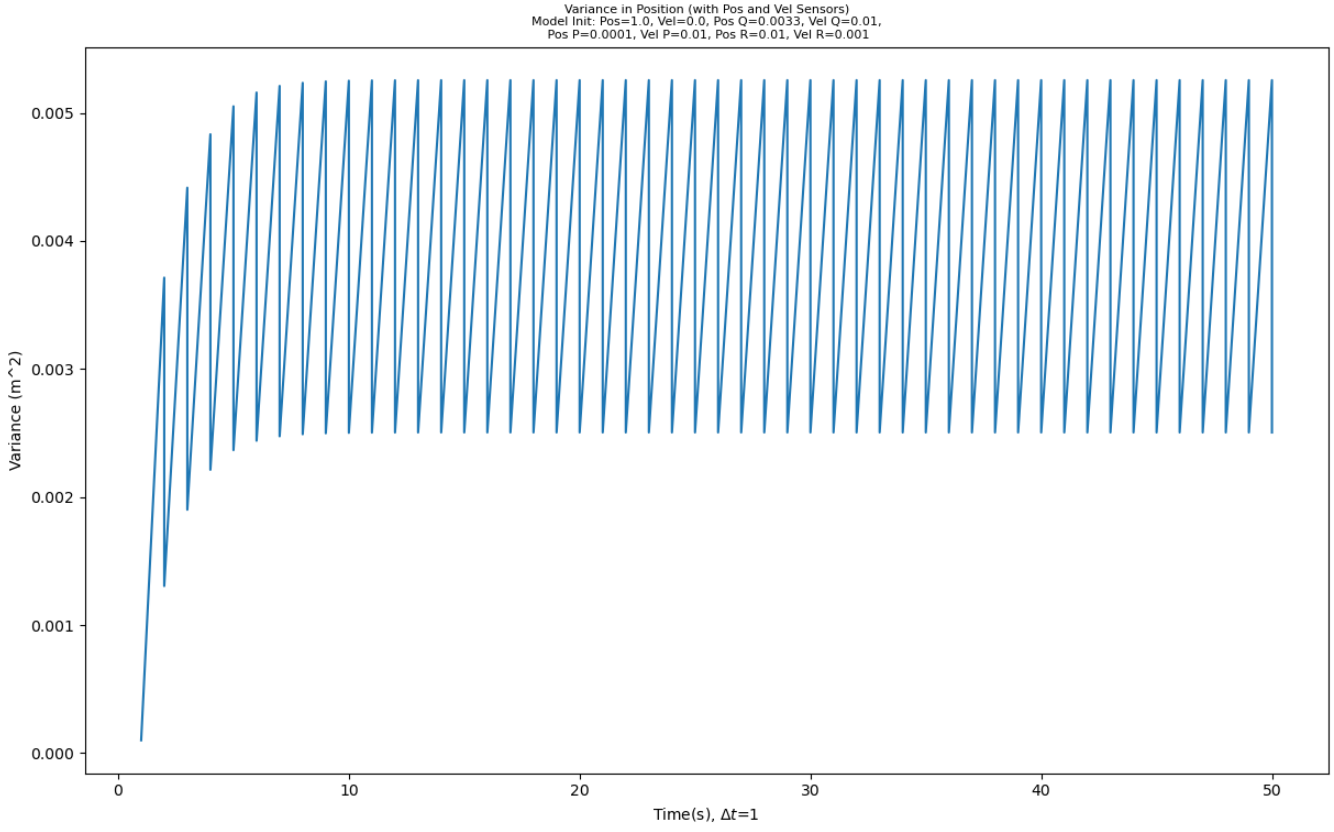


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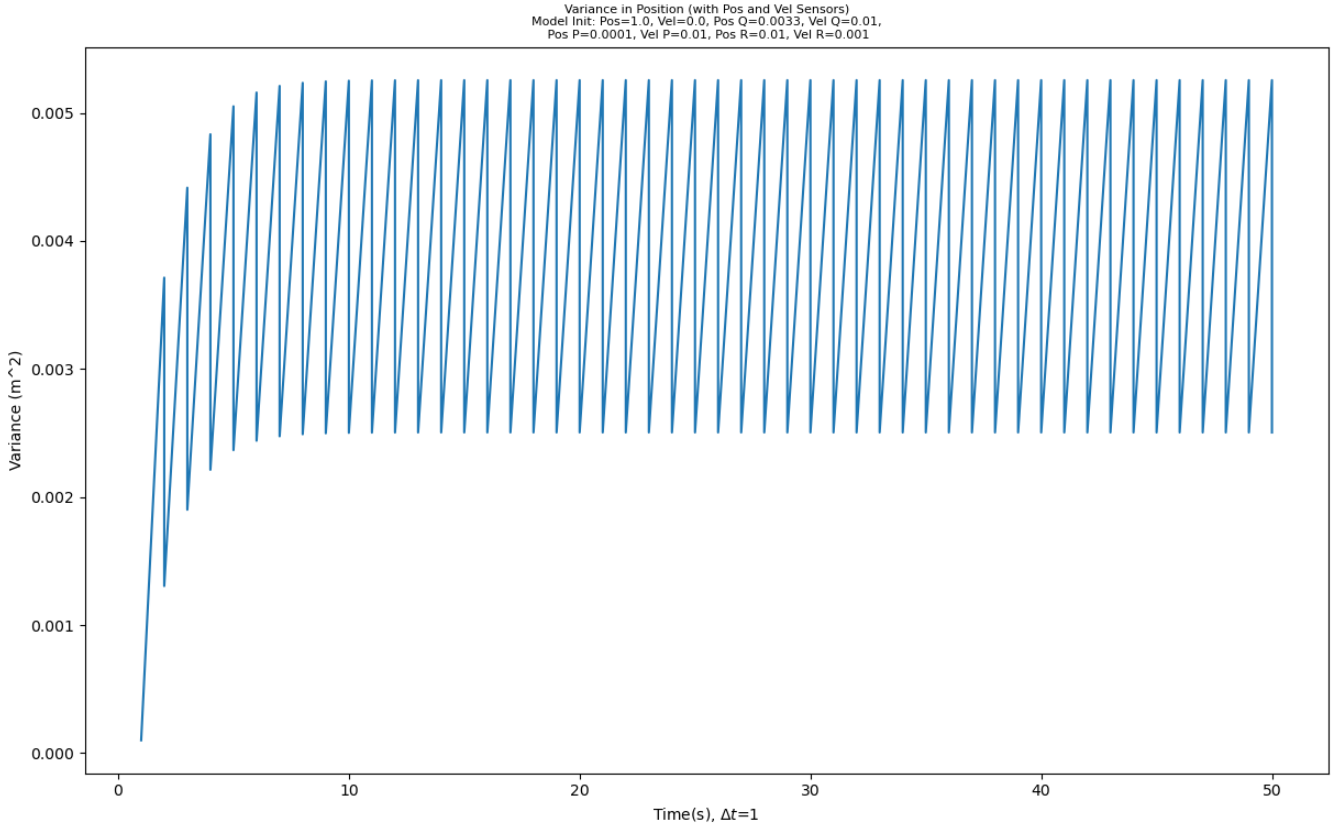


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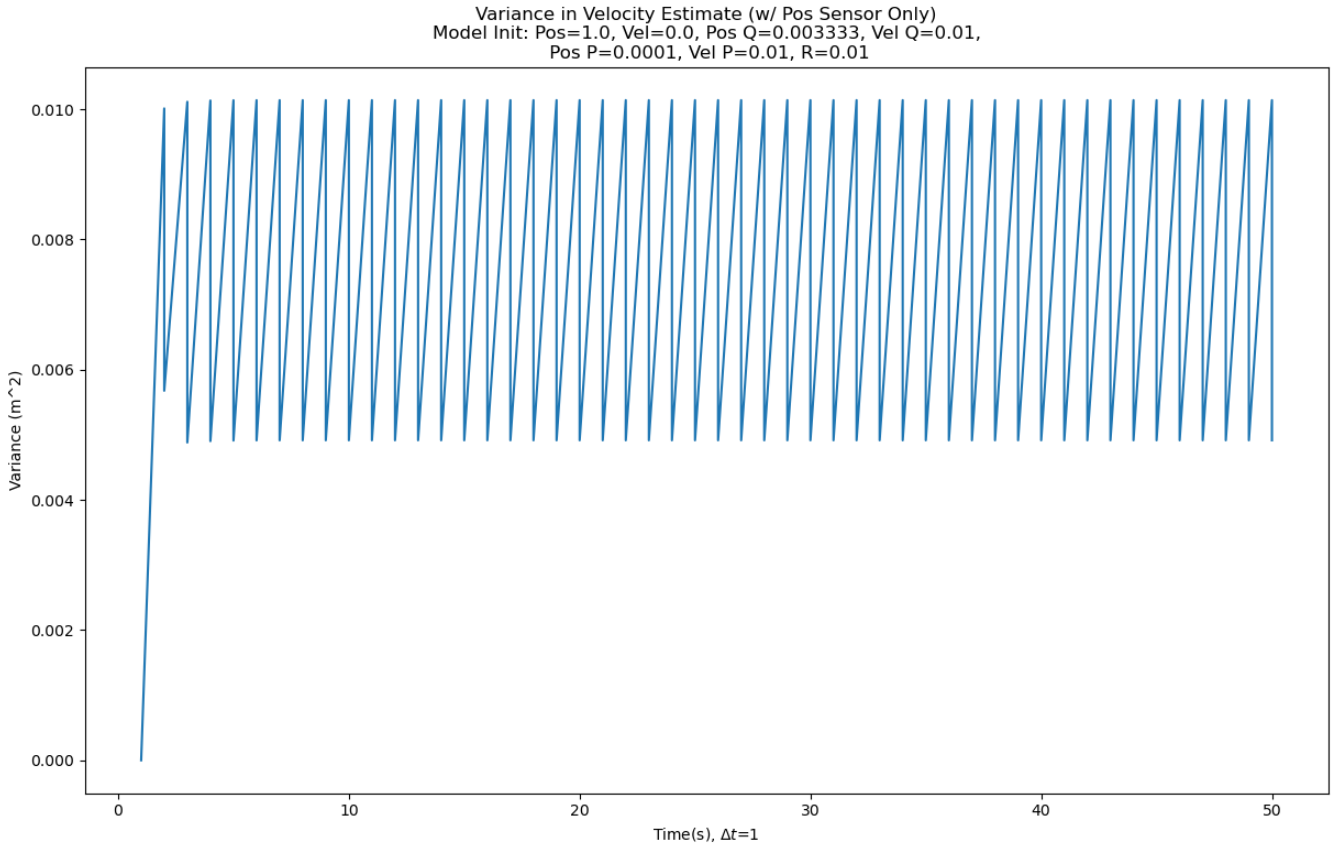


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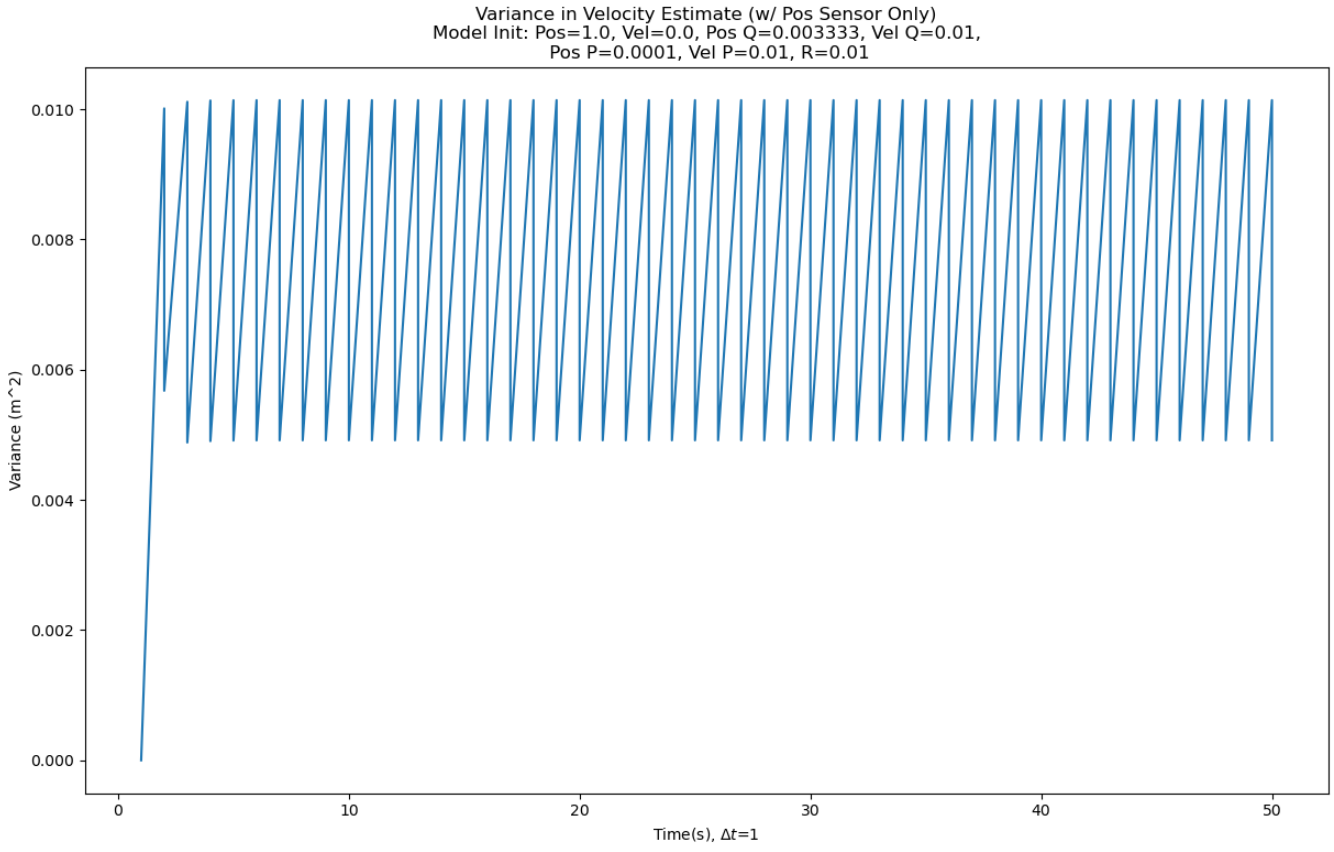


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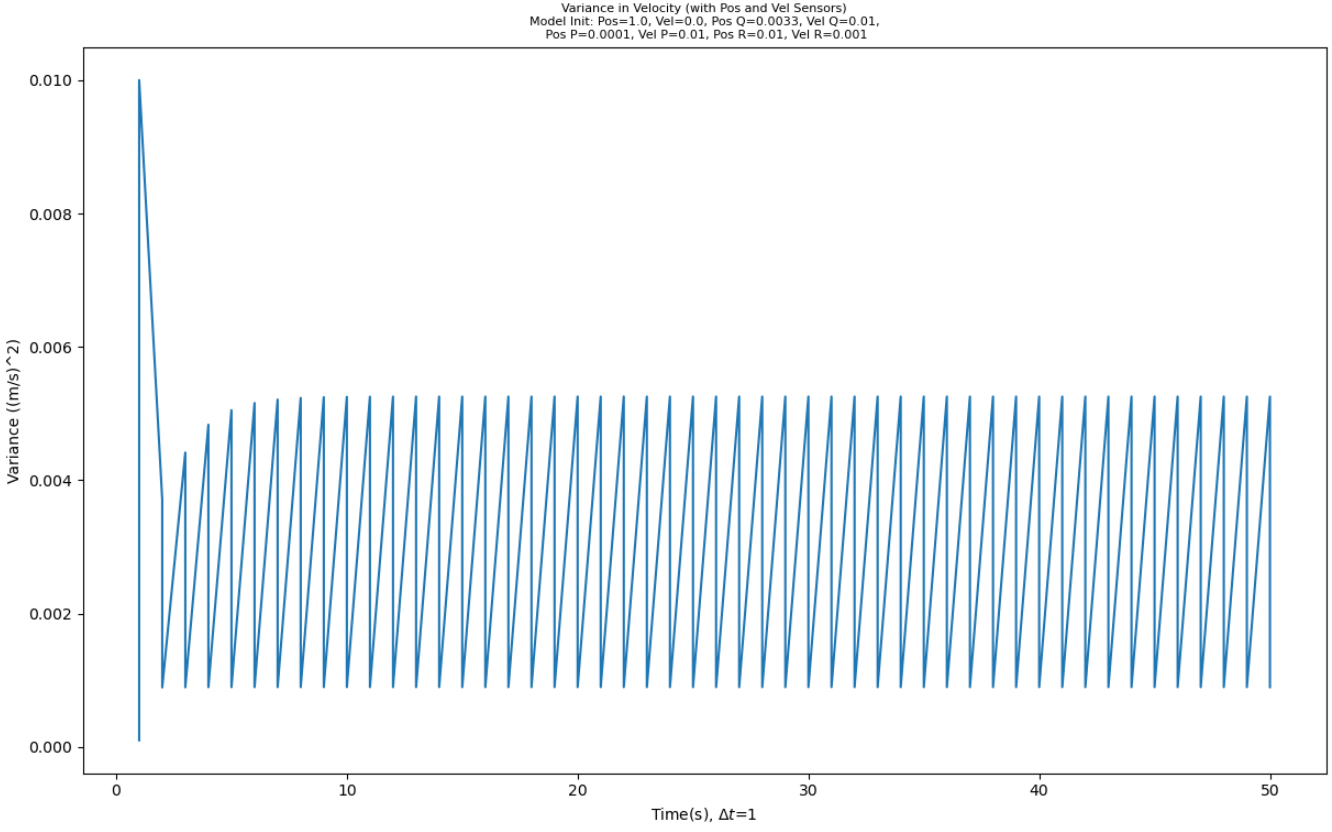


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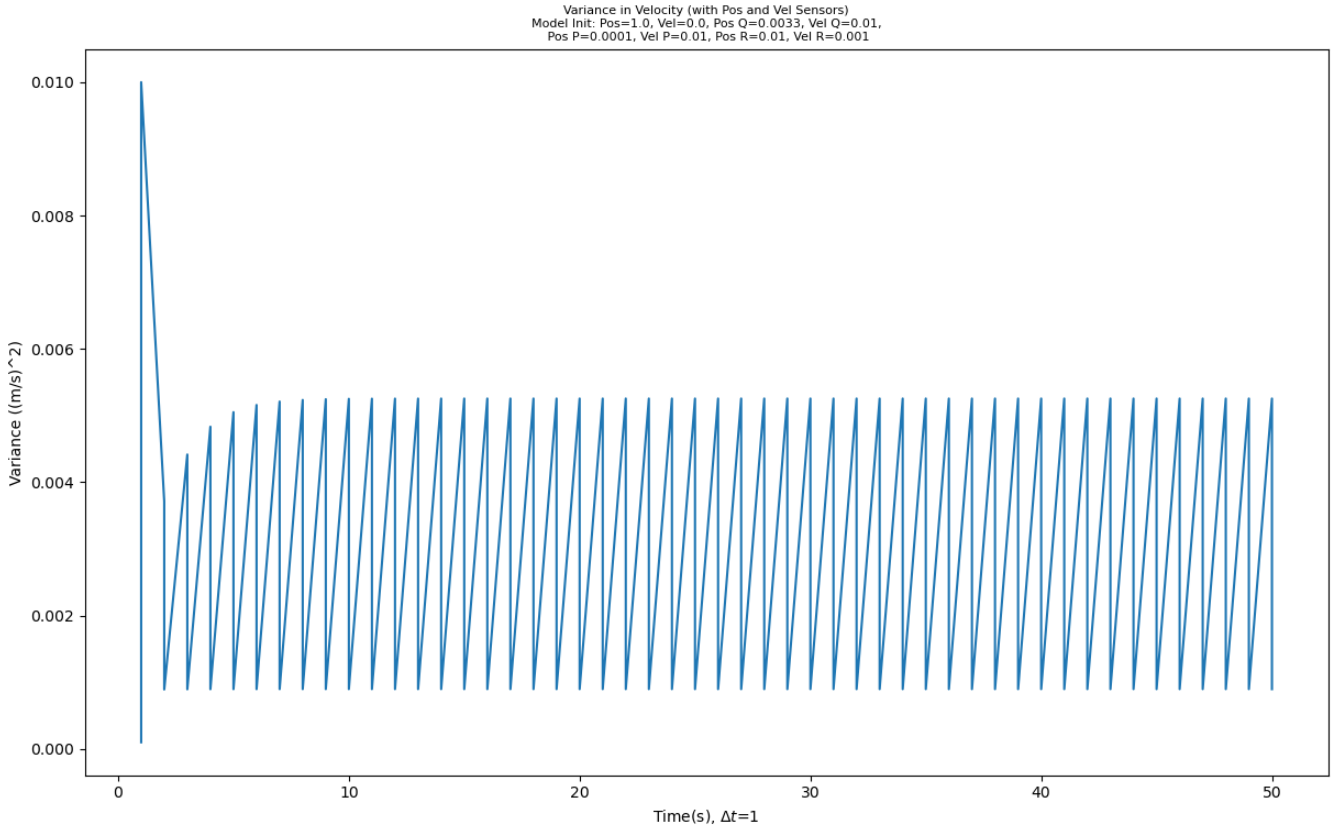


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